

AEMAA Mathematics Applications Test 4 Information

(Term 4: Week 2, Wednesday 23rd October at 8am)

Calculator Assumed, 1 A4 page notes allowed, Formula Sheet Provided.

Response-Test 4 (Univariate Data, Normal Distribution, Applications of trigonometry)	2.1.1 – 2.1.12, 2.2.1 -2.2.2, 2.2.4
--	-------------------------------------

-
- 2.1.1 review the statistical investigation process; identifying a problem and posing a statistical question, collecting or obtaining data, analysing the data, interpreting and communicating the results.
- 2.1.2 classify a categorical variable as ordinal, such as income level (high, medium, low) or nominal, such as place of birth (Australia, overseas) and use tables and bar charts to organise and display data
- 2.1.3 classify a numerical variable as discrete, such as the number of rooms in a house, or continuous, such as the temperature in degrees Celsius
-
- 2.1.5 determine the mean and standard deviation of a data set using technology and use these statistics as measures of location and spread of a data distribution, being aware of their limitations
-
- 2.1.4 with the aid of an appropriate graphical display (chosen from dot plot, stem plot, bar chart or histogram), describe the distribution of a numerical data set in terms of modality (uni or multimodal), shape (symmetric versus positively or negatively skewed), location and spread and outliers, and interpret this information in the context of the data
-
- 2.1.10 construct and use parallel box plots (including the use of the 'Q1 – 1.5 x IQR' and 'Q3 + 1.5 x IQR' criteria for identifying possible outliers) to compare groups in terms of location (median), spread (IQR and range) and outliers, and interpret and communicate the differences observed in the context of the data
-
- 2.1.11 compare groups on a single numerical variable using medians, means, IQRs, ranges or standard deviations, and as appropriate; interpret the differences observed in the context of the data and report the findings in a systematic and concise manner
- 2.1.12 implement the statistical investigation process to answer questions that involve comparing the data for a numerical variable across two or more groups; for example, are Year 11 students the fittest in the school?
-
- 2.1.6 use the number of deviations from the mean (standard scores) to describe deviations from the mean in normally distributed data sets
- 2.1.7 calculate quantiles for normally distributed data with known mean and standard deviation in practical situations
- 2.1.8 use the 68%, 95%, 99.7% rule for data one, two and three standard deviations from the mean in practical situations
- 2.1.9 calculate probabilities for normal distributions with known mean μ and standard deviation σ in practical situations
-
- 2.1.9 calculate probabilities for normal distributions with known mean μ and standard deviation σ in practical situations
- 2.2.1 use trigonometric ratios to determine the length of an unknown side, or the size of an unknown angle in a right-angled triangle
- 2.2.4 solve practical problems involving right-angled and non-right-angled triangles, including problems involving angles of elevation and depression and the use of bearings in navigation
-
- 2.2.2 determine the area of a triangle, given two sides and an included angle by using the rule $area = \frac{1}{2} ab \sin C$, or given three sides by using Heron's rule, and solve related practical problems
-

Equipment required: Ruler Pencil, Eraser, Classpad Calculator.